CHAPTER 6

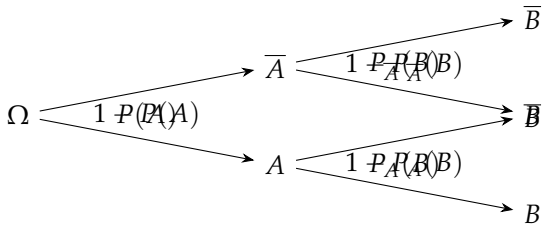
Conditional probability

Skills: model with a tree or a table, calculate, reason about independence, interpret

6.1 Reference course

For two events A and B with $P(A) > 0$, the probability of B given A is

$$P_A(B) = P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$



Key point. In a weighted tree, the probabilities leaving the same node sum to 1. We multiply the probabilities along a path: $P(A \cap B) = P(A)P_A(B)$. We add the probabilities of mutually exclusive paths leading to the event we are looking for.

If $A_1, \ldots, A_n$ form a partition of the sample space, then

$$P(B) = P(A_1)P_{A_1}(B) + \cdots + P(A_n)P_{A_n}(B).$$

Two events A and B with nonzero probabilities are independent if $P(A \cap B) = P(A)P(B)$, or equivalently if $P_A(B) = P(B)$.

Watch out. Do not confuse $P_A(B)$ and $P_B(A)$. The condition comes after the word "given" and is written as a subscript in this notation.

Example 1. 40% of customers buy online. Among them, 12% return their order; among the others, 5% return it. Then $P(R) = 0.40 \times 0.12 + 0.60 \times 0.05 = 0.078$. The overall return rate is therefore 7.8%.

Session 39 — **Conditioning**

Conditional probability — 9 Mar 2027 • B: 2 h – C: 1 h

Objectives.

- Define a conditional probability.
- Move from counts to probabilities.
- Interpret.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 140. 28
2. What is the sum of the branches leaving a node? 1
3. Calculate 0.7×0.4. 0.28
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
5. Simplify $\frac{18}{30}$. $\frac{3}{5}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: define a conditional probability and move from counts to probabilities.

Activity 1. In a survey, 180 customers out of 600 use a premium card and 72 of these 180 customers buy every week. Calculate $P(P)$, $P(P \cap H)$ and $P_P(H)$, then interpret.

Solution. $P(P) = 0.30$, $P(P \cap H) = 0.12$ and $P_P(H) = \frac{72}{180} = 0.40$. Have students verbalise the piece of data that triggers the choice of method.

Written record. For $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. This is the probability of B within the population restricted to A.

Example 2. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $P(P) = 0.30$, $P(P \cap H) = 0.12$ and $P_P(H) = \frac{72}{180} = 0.40$.

Exercise 6.1 · essential.

1. We know $P(A) = 0.35$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

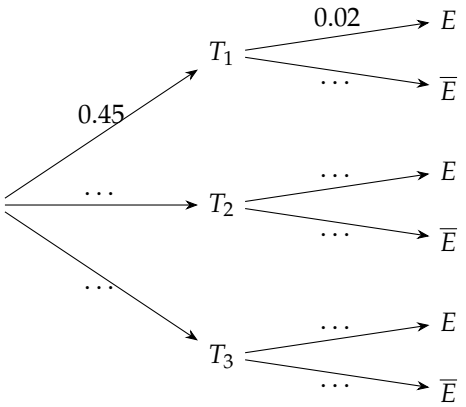
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.35 \times 0.6$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, course pp. 132–137; exercises p. 138, nos. 1–5. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. An online retailer hands its parcels to three carriers: 45% to carrier T_1, 35% to carrier T_2 and the rest to carrier T_3. Customer service finds that 2% of the parcels delivered by T_1 arrive damaged, compared with 5% for T_2 and 8% for T_3.

A parcel is chosen at random. Let T_i be the event "The parcel is delivered by carrier T_i" and E the event "The parcel arrives damaged".



1. Copy and complete the weighted tree above.
2. Calculate the probability that the parcel is delivered by T_3 and arrives damaged.
3. Show that the probability that a parcel arrives damaged is equal to 0.0425.
4. A customer reports a damaged parcel. What is the probability that it was delivered by T_3? Round to 10^{-3}.

Solution. **Target:** Three-branch tree, total and reverse probability.

1. $P(T_3) = 1 - 0.45 - 0.35 = 0.20$. Second-level branches: $P_{T_1}(\overline{E}) = 0.98$; $P_{T_2}(E) = 0.05$ and $P_{T_2}(\overline{E}) = 0.95$; $P_{T_3}(E) = 0.08$ and $P_{T_3}(\overline{E}) = 0.92$.
2. $P(T_3 \cap E) = P(T_3) \times P_{T_3}(E) = 0.20 \times 0.08 = 0.016$.
3. T_1, T_2, T_3 form a partition of the sample space. By the law of total probability: $P(E) = 0.45 \times 0.02 + 0.35 \times 0.05 + 0.20 \times 0.08 = 0.009 + 0.0175 + 0.016 = 0.0425$.
4. $P_E(T_3) = \dfrac{P(T_3 \cap E)}{P(E)} = \dfrac{0.016}{0.0425} \approx 0.376$. Although T_3 carries only 20% of the parcels, it accounts for about 37.6% of the damaged parcels.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 315 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 40 — Weighted trees

Conditional probability

11 Mar 2027 • B: 1 h – C: 2 h

Objectives.

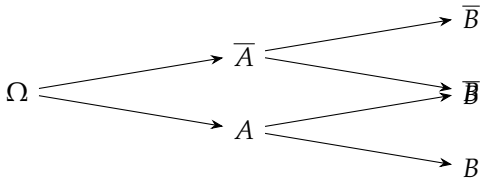
- Construct and complete a tree.
- Read a conditional probability.
- Check the sums.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 25% increase. $1 + \dfrac{25}{100} = 1.25$
2. Calculate 0.8×0.4. 0.32
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
4. Simplify $\dfrac{18}{30}$. $\dfrac{3}{5}$
5. What is the sum of the branches leaving a node? 1

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: construct and complete a tree and read a conditional probability.

Activity 2. 35% of orders are express. Among them, 4% are late; among standard orders, 9% are late. Construct and complete the tree.



Solution. First branches: 0.35 and 0.65. Then 0.04 and 0.96; finally 0.09 and 0.91. Have students verbalise the piece of data that triggers the choice of method.

Written record. In a tree, the branches leaving a node sum to 1. The second-level probabilities are conditional.

Example 3. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. First branches: 0.35 and 0.65. Then 0.04 and 0.96; finally 0.09 and 0.91.

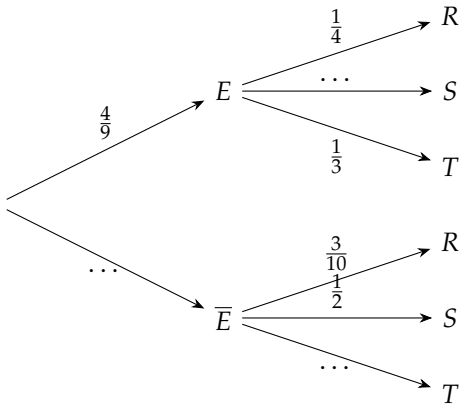
Exercise 6.2 · essential.

1. We know $P(A) = 0.40$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.40 \times 0.2$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, course pp. 133–137; exercises p. 140, nos. 14–18. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. In this exercise, results are given as fractions in lowest terms. Consider the weighted tree below, where E is an event and R, S and T form a partition of the sample space.



1. Complete the tree, justifying each calculation.
2. Calculate $P(E \cap R)$ and $P(\overline{E} \cap S)$.
3. Calculate $P(T)$.

Solution. **Target:** Completing a tree and computing intersections.

1. The branches leaving the same node add up to 1: $P(\overline{E}) = 1 - \dfrac{4}{9} = \dfrac{5}{9}$; $P_E(S) = 1 - \dfrac{1}{4} - \dfrac{1}{3} = \dfrac{5}{12}$; $P_{\overline{E}}(T) = 1 - \dfrac{3}{10} - \dfrac{1}{2} = \dfrac{1}{5}$.
2. $P(E \cap R) = \dfrac{4}{9} \times \dfrac{1}{4} = \dfrac{1}{9}$ and $P(\overline{E} \cap S) = \dfrac{5}{9} \times \dfrac{1}{2} = \dfrac{5}{18}$.
3. E and $\overline{E}$ form a partition, so $P(T) = P(E \cap T) + P(\overline{E} \cap T) = \dfrac{4}{9} \times \dfrac{1}{3} + \dfrac{5}{9} \times \dfrac{1}{5} = \dfrac{4}{27} + \dfrac{1}{9} = \dfrac{4}{27} + \dfrac{3}{27} = \dfrac{7}{27}$.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 320 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 41 — Product of probabilities along a path

Conditional probability

16 Mar 2027 • B: 2 h – C: 1 h

Objectives.

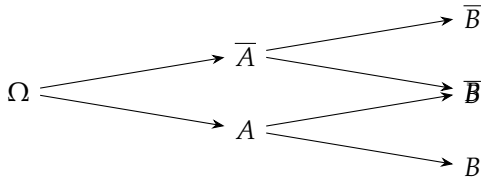
- Calculate an intersection.
- Combine paths.
- Translate an event.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 9 - x + 2$. $2x + 11$
2. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
3. Simplify $\dfrac{18}{30}$. $\dfrac{3}{5}$
4. What is the sum of the branches leaving a node? 1
5. Calculate 0.9×0.4. 0.36

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: calculate an intersection and combine paths.

Activity 3. Using the tree from session 40, calculate the probability of an order being express and late, then the total probability of a delay. Interpret both results.



Solution. $P(E \cap R) = 0.35 \times 0.04 = 0.014$. $P(R) = 0.014 + 0.65 \times 0.09 = 0.0725$. Have students verbalise the piece of data that triggers the choice of method.

Written record. We multiply along a path: $P(A \cap B) = P(A)P_A(B)$. We add the mutually exclusive paths that realise the event.

Example 4. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $P(E \cap R) = 0.35 \times 0.04 = 0.014$. $P(R) = 0.014 + 0.65 \times 0.09 = 0.0725$.

Exercise 6.3 · essential.

1. We know $P(A) = 0.45$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.45 \times 0.3$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, p. 140, nos. 14–18. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. In a gym, 65% of members have an annual membership, the others a monthly membership. Each member chooses a main activity. Among annual members, 50% choose weight training, 30% group classes and the rest swimming. Among monthly members, 40% choose weight training, 45% group classes and the rest swimming.

A member is chosen at random. Let A: "The member has an annual membership", M: "Their main activity is weight training", C: "Their main activity is a group class" and N: "Their main activity is swimming".

1. Draw a weighted tree describing the situation.
2. Describe the event $\overline{A} \cap C$ in a sentence, then calculate its probability.
3. Calculate the probability that the chosen member's main activity is swimming.
4. The chosen member mainly swims. Calculate the probability that they have an annual membership. Round to 10^{-3}.

Solution. **Target:** Translating an event, product along a path.

1. First level: $P(A) = 0.65$ and $P(\overline{A}) = 0.35$. Under A: $P_A(M) = 0.50$, $P_A(C) = 0.30$, $P_A(N) = 1 - 0.50 - 0.30 = 0.20$. Under $\overline{A}$: $P_{\overline{A}}(M) = 0.40$, $P_{\overline{A}}(C) = 0.45$, $P_{\overline{A}}(N) = 1 - 0.40 - 0.45 = 0.15$.
2. $\overline{A} \cap C$: "The member has a monthly membership and their main activity is a group class". $P(\overline{A} \cap C) = 0.35 \times 0.45 = 0.1575$.
3. By the law of total probability (A, $\overline{A}$ form a partition): $P(N) = 0.65 \times 0.20 + 0.35 \times 0.15 = 0.13 + 0.0525 = 0.1825$.
4. $P_N(A) = \dfrac{P(A \cap N)}{P(N)} = \dfrac{0.13}{0.1825} \approx 0.712$.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 325 and the rate or ratio with 7. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 42 — Independence

Conditional probability

18 Mar 2027 • B: 1 h – C: 2 h

Objectives.

- Test independence.
- Compare probabilities.
- Justify a conclusion.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 5% of 60. 3
2. Simplify $\frac{18}{30}$. $\frac{3}{5}$
3. What is the sum of the branches leaving a node? 1
4. Calculate 0.3×0.4. 0.12
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: test independence and compare probabilities.

Activity 4. In a population, $P(A) = 0.4$, $P(B) = 0.3$ and $P(A \cap B) = 0.12$. Test independence in two ways. Then change $P(A \cap B)$ to 0.15 and start again.

Solution. $0.4 \times 0.3 = 0.12$: independence. With 0.15, the events are no longer independent. Have students verbalise the piece of data that triggers the choice of method.

Written record. A and B are independent if $P(A \cap B) = P(A)P(B)$, equivalently if $P_A(B) = P(B)$ when $P(A) > 0$.

Example 5. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $0.4 \times 0.3 = 0.12$: independence. With 0.15, the events are no longer independent.

Exercise 6.4 · essential.

1. We know $P(A) = 0.20$, $P_A(B) = 0.4$ and $P_{\overline{A}}(B) = 0.6$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.20 \times 0.4$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, course pp. 134–137; exercises p. 144, nos. 33–35. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. In a call centre, a call is chosen at random. Let A be the event "The customer waits more than 5 minutes before being answered" and B the event "The call is cut off because of a network incident". We know that $P(A) = 0.12$ and $P(B) = 0.05$. In questions 1 to 3, assume that A and B are independent.

1. Calculate the probability that the customer waits more than 5 minutes and that the call is cut off.
2. Calculate the probability that neither of these two incidents occurs.
3. Deduce the probability that at least one of the two incidents occurs.
4. An audit actually shows that, among calls with a wait of more than 5 minutes, 9% are cut off. Is the independence assumption confirmed? Justify.

Solution. **Target:** Using and testing independence.

1. By independence, $P(A \cap B) = P(A) \times P(B) = 0.12 \times 0.05 = 0.006$.
2. If A and B are independent, so are $\overline{A}$ and $\overline{B}$: $P(\overline{A} \cap \overline{B}) = 0.88 \times 0.95 = 0.836$.
3. The event "at least one incident" is the complement of "no incident": its probability is $1 - 0.836 = 0.164$.
4. The audit gives $P_A(B) = 0.09$, whereas $P(B) = 0.05$. Since $P_A(B) \neq P(B)$, A and B are not independent: a long wait goes together with a cut-off more often.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 330 and the rate or ratio with 2. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Notes

Session 43 — Partition and total probability

Conditional probability

23 Mar 2027 • B: 2 h – C: 1 h

Objectives.

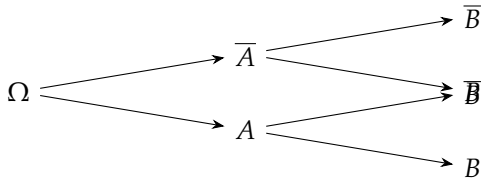
- Recognise a partition.
- Apply total probability.
- Organise the calculation.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 10% increase. $1 + \dfrac{10}{100} = 1.10$
2. What is the sum of the branches leaving a node? 1
3. Calculate 0.4×0.4. 0.16
4. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
5. Simplify $\dfrac{18}{30}$. $\dfrac{3}{5}$

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: recognise a partition and apply total probability.

Activity 5. A factory has three lines A, B, C producing 20%, 50% and 30% of the items respectively. Their defect rates are 1%, 2% and 4%. Calculate the overall defect rate.



Solution. $P(D) = 0.2 \times 0.01 + 0.5 \times 0.02 + 0.3 \times 0.04 = 0.024$, that is 2.4%. Have students verbalise the piece of data that triggers the choice of method.

Written record. If (A_i) is a partition, $P(B) = \sum_i P(A_i) P_{A_i}(B)$. Each term corresponds to a path leading to B.

Example 6. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $P(D) = 0.2 \times 0.01 + 0.5 \times 0.02 + 0.3 \times 0.04 = 0.024$, that is 2.4%.

Exercise 6.5 · essential.

1. We know $P(A) = 0.25$, $P_A(B) = 0.5$ and $P_{\overline{A}}(B) = 0.7$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.25 \times 0.5$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, p. 138, nos. 1–5, then p. 142, nos. 22–25. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A group employs 1,600 staff across three sites: the head office, the factory and the warehouse. The staff who work from home at least one day a week have been counted.

	Remote work	No remote work	Total
Head office	300	···	400
Factory	···	···	···
Warehouse	···	340	400
Total	420	···	1,600

1. Complete the table.
2. An employee is chosen at random. Let S_1: "They work at the head office", S_2: "They work at the factory", S_3: "They work at the warehouse" and T: "They work from home". Draw the weighted tree whose first level is S_1, S_2, S_3 and second level is T, $\overline{T}$.
3. Draw the weighted tree whose first level is T, $\overline{T}$ and second level is S_1, S_2, S_3. Give the probabilities as fractions in lowest terms, then rounded to 10^{-3}.

Solution. **Target:** From a two-way table to weighted trees.

1. Head office: $400 - 300 = 100$ without remote work. Warehouse: $400 - 340 = 60$ with remote work. Factory: $1{,}600 - 400 - 400 = 800$ staff, of whom $420 - 300 - 60 = 60$ work from home and $800 - 60 = 740$ do not. Total without remote work: $100 + 740 + 340 = 1{,}180$ $(= 1{,}600 - 420)$.
2. $P(S_1) = \dfrac{400}{1{,}600} = 0.25$, $P(S_2) = \dfrac{800}{1{,}600} = 0.5$, $P(S_3) = 0.25$.
   $P_{S_1}(T) = \dfrac{300}{400} = 0.75$ and $P_{S_1}(\overline{T}) = 0.25$; $P_{S_2}(T) = \dfrac{60}{800} = 0.075$ and $P_{S_2}(\overline{T}) = 0.925$; $P_{S_3}(T) = \dfrac{60}{400} = 0.15$ and $P_{S_3}(\overline{T}) = 0.85$.
3. $P(T) = \dfrac{420}{1{,}600} = \dfrac{21}{80} = 0.2625$ and $P(\overline{T}) = \dfrac{59}{80} = 0.7375$.
   $P_T(S_1) = \dfrac{300}{420} = \dfrac{5}{7} \approx 0.714$, $P_T(S_2) = \dfrac{60}{420} = \dfrac{1}{7} \approx 0.143$, $P_T(S_3) = \dfrac{1}{7} \approx 0.143$.
   $P_{\overline{T}}(S_1) = \dfrac{100}{1{,}180} = \dfrac{5}{59} \approx 0.085$, $P_{\overline{T}}(S_2) = \dfrac{740}{1{,}180} = \dfrac{37}{59} \approx 0.627$, $P_{\overline{T}}(S_3) = \dfrac{340}{1{,}180} = \dfrac{17}{59} \approx$

0.288.
Check: the fractions leaving each node add up to 1.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 335 and the rate or ratio with 3. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 44 — **Problems in context**

Conditional probability — 25 Mar 2027 • B: 1 h – C: 2 h

Objectives.

- Choose a tree or a table.
- Solve a problem in context.
- Interpret.

Fluency · 6 to 8 minutes, no calculator.

1. Simplify $3x + 5 - x + 2$. $2x + 7$
2. Calculate 0.5×0.4. 0.20
3. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
4. Simplify $\dfrac{18}{30}$. $\dfrac{3}{5}$
5. What is the sum of the branches leaving a node? 1

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: choose a tree or a table and solve a problem in context.

Activity 6. Turn a table of counts (customers/subscription/cancellation) into a tree, then answer a question in each direction: probability of cancelling given the subscription; probability of having the subscription given the cancellation.

Solution. The first reading uses a row or a branch; the second requires dividing the intersection by the total number of cancellations. Have students verbalise the piece of data that triggers the choice of method.

Written record. Choosing between a tree and a table: the tree highlights the order of conditioning; the table suits counts and reversed readings well.

Example 7. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The first reading uses a row or a branch; the second requires dividing the intersection by the total number of cancellations.

Exercise 6.6 · essential.

1. We know $P(A) = 0.30$, $P_A(B) = 0.6$ and $P_{\overline{A}}(B) = 0.8$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.

3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.30 \times 0.6$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, p. 142, nos. 22–25, then p. 144, nos. 30–32. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A DIY chain surveys its customers. 45% of them have downloaded the chain's loyalty app; among these, 60% made at least one purchase on special offer during the last month. Overall, 36% of the customers surveyed made at least one purchase on special offer during the last month.

A surveyed customer is chosen at random. Let A: "The customer has downloaded the app" and R: "The customer made at least one purchase on special offer last month".

1. Give $P(A)$, $P(R)$ and $P_A(R)$.
2. Calculate $P(A \cap R)$, then $P_R(A)$. Interpret the latter result.
3. Calculate the probability that a customer who has not downloaded the app made a purchase on special offer. Round to 10^{-3} and compare with $P_A(R)$.

Solution. **Target:** Reversing a conditional probability.

1. $P(A) = 0.45$, $P(R) = 0.36$ and $P_A(R) = 0.60$.
2. $P(A \cap R) = P(A) \times P_A(R) = 0.45 \times 0.60 = 0.27$. Then $P_R(A) = \dfrac{P(A \cap R)}{P(R)} = \dfrac{0.27}{0.36} = 0.75$: three quarters of the customers who bought on special offer have downloaded the app.
3. $P(\overline{A} \cap R) = P(R) - P(A \cap R) = 0.36 - 0.27 = 0.09$ and $P(\overline{A}) = 0.55$, so $P_{\overline{A}}(R) = \dfrac{0.09}{0.55} \approx 0.164$. This is much less than $P_A(R) = 0.60$: customers with the app buy on special offer far more often.

Homework. Redo practice exercise 3 with the following data: replace the initial value with 340 and the rate or ratio with 4. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion).

> Ask fast students to produce a counterexample, a justification or a second method.

Session 45 — Mixed practice

Conditional probability

30 Mar 2027 • B: 2 h – C: 1 h

Objectives.

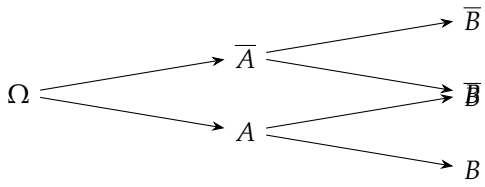
- Calculate a reverse probability.
- Chain total probability and conditioning.
- Check the result.

Fluency · 6 to 8 minutes, no calculator.

1. Calculate 20% of 120. 24
2. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63
3. Simplify $\frac{18}{30}$. $\frac{3}{5}$
4. What is the sum of the branches leaving a node? 1
5. Calculate 0.6×0.4. 0.24

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: calculate a reverse probability and chain total probability and conditioning.

Activity 7. In the situation of session 43, given that an item is defective, calculate the probability that it comes from line C. Explain why this result is not 30%.



Solution. $P_D(C) = \dfrac{0.3 \times 0.04}{0.024} = 0.5$. Conditioning changes the distribution because the defect rates differ. Have students verbalise the piece of data that triggers the choice of method.

Written record. For a reverse probability, first calculate the total probability of the conditioning event, then use the definition of conditional probability.

Example 8. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. $P_D(C) = \dfrac{0.3 \times 0.04}{0.024} = 0.5$. Conditioning changes the distribution because the defect rates differ.

Exercise 6.7 · essential.

1. We know $P(A) = 0.35$, $P_A(B) = 0.2$ and $P_{\overline{A}}(B) = 0.4$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.35 \times 0.2$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, pp. 144–146. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. In this exercise, results are given as fractions in lowest terms. During a promotional event, a customer draws one ticket at random from ten tickets numbered 1 to 10.

- If the number is 1 or 2, they scratch a card taken at random from box X, which contains 3 winning cards out of 8.
- If the number is a multiple of 3, they scratch a card from box Y, which contains 2 winning cards out of 9.
- Otherwise, they scratch a card from box Z, which contains 1 winning card out of 6.

Let X, Y, Z be the events "The card comes from box X" (Y, Z respectively) and G the event "The card is a winning card".

1. Justify that $P(X) = \dfrac{1}{5}$, $P(Y) = \dfrac{3}{10}$ and $P(Z) = \dfrac{1}{2}$, then represent the situation with a weighted tree.
2. Prove that the probability that the card is a winning card is equal to $\dfrac{9}{40}$.
3. The customer has won. What is the probability that their card came from box X?

Solution. **Target:** Two-stage experiment, fractional results.

1. The ten numbers are equally likely. X corresponds to $\{1,2\}$: $P(X) = \dfrac{2}{10} = \dfrac{1}{5}$. Y corresponds to $\{3,6,9\}$: $P(Y) = \dfrac{3}{10}$. Z corresponds to the five other numbers $\{4,5,7,8,10\}$: $P(Z) = \dfrac{5}{10} = \dfrac{1}{2}$.
   Tree: under X, $P_X(G) = \dfrac{3}{8}$ and $P_X(\overline{G}) = \dfrac{5}{8}$; under Y, $P_Y(G) = \dfrac{2}{9}$ and $P_Y(\overline{G}) = \dfrac{7}{9}$; under Z, $P_Z(G) = \dfrac{1}{6}$ and $P_Z(\overline{G}) = \dfrac{5}{6}$.

2. X, Y, Z form a partition: $P(G) = \frac{1}{5} \times \frac{3}{8} + \frac{3}{10} \times \frac{2}{9} + \frac{1}{2} \times \frac{1}{6} = \frac{3}{40} + \frac{1}{15} + \frac{1}{12} = \frac{9}{120} + \frac{8}{120} + \frac{10}{120} = \frac{27}{120} = \frac{9}{40}$.
3. $P_G(X) = \frac{P(X \cap G)}{P(G)} = \frac{3/40}{9/40} = \frac{1}{3}$.

Homework. Redo practice exercise 2 with the following data: replace the initial value with 345 and the rate or ratio with 5. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1, 2 and 3.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.

Session 46 — Review and assessment

Conditional probability

1 Apr 2027 • B: 1 h – C: 2 h

Objectives.

- Use all the methods.
- Write up with the notation.
- Remedy mistakes.

Fluency · 6 to 8 minutes, no calculator.

1. Write the multiplier for a 25% increase. $1 + \dfrac{25}{100} = 1.25$
2. Simplify $\dfrac{18}{30}$. $\dfrac{3}{5}$
3. What is the sum of the branches leaving a node? 1
4. Calculate 0.7×0.4. 0.28
5. If $P(A) = 0.37$, calculate $P(\overline{A})$. 0.63

Prerequisites. Reread the last written record and be able to carry out the basic calculations used today. Before the investigation, have the keywords restated: use all the methods and write up with the notation.

Activity 8. Complete a four-station circuit: complete a tree; calculate an intersection; apply the total probability formula; test independence. Each group leaves a written justification for the next group.

Solution. The work produced must contain formulas and interpretations. Mistakes are sorted into: notation, complement, product, sum, reversed conditioning. Have students verbalise the piece of data that triggers the choice of method.

Written record. Always name the events, complete all the branches, write the formula before calculating and interpret the final probability.

Example 9. Take the data of the investigation and write the solution in order: definition of the objects, property used, exact or approximate calculation, then a sentence giving the answer. The model answer appears in the teacher edition.

Solution. The work produced must contain formulas and interpretations. Mistakes are sorted into: notation, complement, product, sum, reversed conditioning.

Exercise 6.8 · essential.

1. We know $P(A) = 0.40$, $P_A(B) = 0.3$ and $P_{\overline{A}}(B) = 0.5$. Construct the tree.
2. Calculate $P(A \cap B)$ then $P(B)$.
3. Calculate $P_B(A)$ and test independence.

Solution. $P(A \cap B) = 0.40 \times 0.3$. Then $P(B) = P(A)P_A(B) + (1 - P(A))P_{\overline{A}}(B)$ and $P_B(A) = \dfrac{P(A \cap B)}{P(B)}$. Compare with $P(A)$.

Further practice: Calao, pp. 146–147. The application exercise below is original; these references point to textbook exercises for extra practice and for its figures.

Application. A business incubator selects the projects it supports in three stages. 30% of the applications submitted pass the pre-selection. Pre-selected projects are presented to a panel, which keeps 50% of them. Finally, an investors' committee funds 20% of the projects kept by the panel. A project eliminated at one stage does not go on to the following stages.

An application is chosen at random. Let P: "The application passes the pre-selection", J: "The project is kept by the panel" and F: "The project is funded".

1. Draw a weighted tree describing the situation. Prove that the probability that a project is not funded is equal to 0.97.
2. Eight entrepreneurs submit an application; assume that the applications are examined independently of one another. Calculate the probability that at least one of the eight projects is funded. Round to 10^{-3}.
3. What is the minimum number of applications the incubator must receive for the probability of funding at least one project to exceed 0.99?

Solution. **Target:** Review: path, complement and independent repetition.

1. Tree: P (0.30) or $\overline{P}$ (0.70, end); after P: J (0.50) or $\overline{J}$ (0.50, end); after J: F (0.20) or $\overline{F}$ (0.80). A project is funded only if it passes all three stages, so, multiplying along this path: $P(F) = 0.30 \times 0.50 \times 0.20 = 0.03$ and $P(\overline{F}) = 1 - 0.03 = 0.97$.
2. By independence, the probability that none of the eight projects is funded is 0.97^8. The required probability is $1 - 0.97^8 \approx 0.216$.
3. We look for the smallest whole number n such that $1 - 0.97^n > 0.99$, i.e. $0.97^n < 0.01$. Since log is increasing, this is equivalent to $n \log(0.97) < \log(0.01)$, i.e. $n > \dfrac{\log(0.01)}{\log(0.97)} \approx 151.2$ (dividing by $\log(0.97) < 0$ reverses the inequality). At least 152 applications are needed (check: $1 - 0.97^{151} \approx 0.98994$ and $1 - 0.97^{152} \approx 0.99024$).

Homework. Redo practice exercise 3 with the following data: replace the initial value with 350 and the rate or ratio with 6. Write it up in four lines: data, formula, calculation, conclusion.

Solution. The solution depends on the exercise redone. Check above all the consistency of the model, the number of iterations, the rounding and the concluding sentence.

Exit ticket · 3 minutes. On a half-sheet: write the main rule of the session, work through a very short example and indicate your confidence level from 1 to 4. Key question(s) of the day: 1 and 2.

Method. **One-hour plan:** 7 min of fluency; 10 min of pooling and written record; 28 min of essential exercises; 5 min of correction; 3 min of exit ticket.
Two-hour plan: add 20 min of investigation, 20 min of formalisation, a second independent exercise and 10 min of comparative correction.
Differentiation: give struggling students a four-box sheet (data, model, calculation, conclusion). Ask fast students to produce a counterexample, a justification or a second method.